

HPS Gaussian-Process Resonance Search

Analysis Note v4.9.13:

Background Profiling and Fixed-Background Tests

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Conditional methodology study. This note extends the v4.9.12 observed results and v4.9.12.5 injection diagnostics. It compares background treatments and adds 27,000 pointwise pseudoexperiments. The fixed-background curves assume the estimated GP mean is exactly known. The 2016 results and their combination retain the disclosed numerical exception. These tests do not certify unconditional coverage or global significance.

1 Principal findings and scope

The use of correlated background nuisance parameters in a Poisson likelihood has a sound literature basis. Profiling the latent log-background directly provides a positive-background alternative: its 2021 limits differ from the released Gaussian approximation by at most 1.189%, and by only 0.232% when both models use the same numerical solver. Changing this distribution does not remove the peak–dip structure.

Fixing the GP mean produces narrower conditional limits, but the injection tests identify the price. It recovers signals with appropriate errors when that mean is the true, known background. With GP uncertainty included in generation, the fixed method excludes the stronger true injection in 22–32% of toys at nominal 90% CL_s . Refitting sidebands exposes bias in both methods: at 71 MeV the corresponding fractions are 69.8% fixed and 29.0% profiled. This is a limitation of the current procedure as well as a reason not to adopt the uncorrected fixed-background result.

Dataset	Observed grid (MeV)	GP support (MeV)	Length ceiling factor
2015, 100%	19–90	14–135	8
2016, 100%	39–180	30–210	12
2021, 10%	50–250	36–300	15
All three, shared ϵ^2	50–90	Constituent supports	Constituent factors

Table 1: Frozen v4.9.12 inputs. Observed hypotheses are spaced by 1 MeV. The nominal resolution, bin-integrated signal templates, moving $\pm 2.25\sigma_m$ exclusion/fit mask, and yield conversion are retained. The length ceiling factors multiply the dataset’s prescribed resolution scale.

The observed extension contains 456 scope/mass points. The extraction tests use only 2021 at six declared masses and three signal strengths. No new 2015/2016 injection calibration or scan-wide toy ensemble is inferred from them.

2 Likelihood construction and literature basis

Let n_i be the counts in the signal fit window, w_i its normalized signal template ($\sum_i w_i = 1$), and A its signal yield. The sideband-trained log-GP posterior has mean g and covariance K . Its count mean and covariance before numerical conditioning are

$$b_i = \exp(g_i + K_{ii}/2), \quad C_{ij} = b_i b_j [\exp(K_{ij}) - 1]. \quad (1)$$

The release approximates that count distribution by correlated Gaussian nuisances. With LL^T the effective conditioned count covariance, its objective is

$$\ell(A, \theta) = \sum_i \left[\lambda_i - n_i + n_i \log \left(\frac{n_i}{\lambda_i} \right) \right] + \frac{1}{2} \theta^T \theta, \quad \lambda = b + L\theta + Aw. \quad (2)$$

The $n_i = 0$ term is interpreted by continuity and $\lambda_i > 0$ is required. Centering the Poisson objective removes only a data-dependent constant. Correlated background modes are profiled at every tested A ; the reviewed kernel hyperparameters remain frozen.

The controlled alternative uses the same Poisson objective and penalty but profiles a latent positive log-intensity,

$$\lambda_i = \exp[g_i + (R\theta)_i] + Aw_i, \quad RR^T = K_{\text{conditioned}}. \quad (3)$$

Its zero-nuisance Asimov background is $\exp(g)$, whereas the Gaussian and fixed models use b . The largest mean/median difference is 0.000472%. The fixed reference sets $L\theta = 0$ in Eq. (2); its estimate b is treated as known.

Frate et al. connect GP backgrounds with constrained Poisson intensity models and positive log-intensity formulations [1]. The regularization or posterior constraint must still be distinguished from an independently measured frequentist auxiliary likelihood. Cowan et al. provide the profile-likelihood and bounded asymptotic framework [2]; neither reference establishes coverage for this particular fitted background, moving mask, or selected kernel configuration.

All upper limits solve the frozen bounded, piecewise-asymptotic $CL_s(A) = 0.1$ construction. The likelihood-ratio denominator is the signed best fit when $\hat{A} \geq 0$ and the null fit otherwise; the test statistic vanishes for $\hat{A} > A$. Background-only Asimov profiles set the asymptotic scale, including the deficit branch. For local discovery diagnostics,

$$r = \text{sgn}(\hat{A}) \sqrt{2[\ell(0, \hat{\theta}_0) - \ell(\hat{A}, \hat{\theta})]}, \quad Z = \max(r, 0), \quad p_0 = 1 - \Phi(Z). \quad (4)$$

Thus deficits have $p_0 = 0.5$ under this reporting convention. These are fixed-mass asymptotic values; no trials correction is applied.

The maximum marginal GP standard deviation is 0.307% of the mean. The two profile distributions are consequently close, although their uncertainty can exceed the counting error in these high-statistics bins. Independent optimizer checks support the stable solutions; the released columns remain the comparison reference.

3 Controlled 2021 comparison

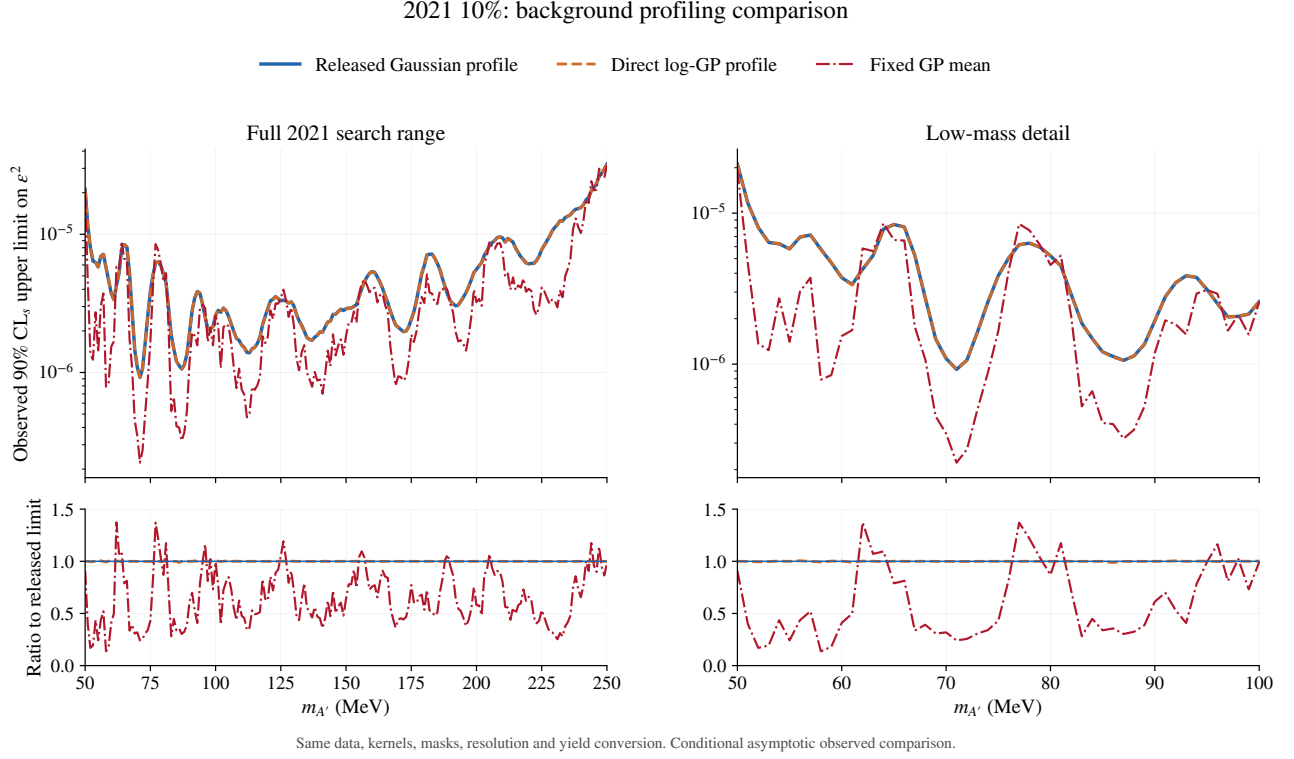


Figure 1: Observed 2021 10% limits under the released Gaussian profile, the direct log-GP profile, and the fixed GP mean. The lower panels divide by the released limit. Every curve uses the same native data and frozen analysis settings; the standard dimuon branching correction is applied once above threshold. The fixed curve omits estimation uncertainty.

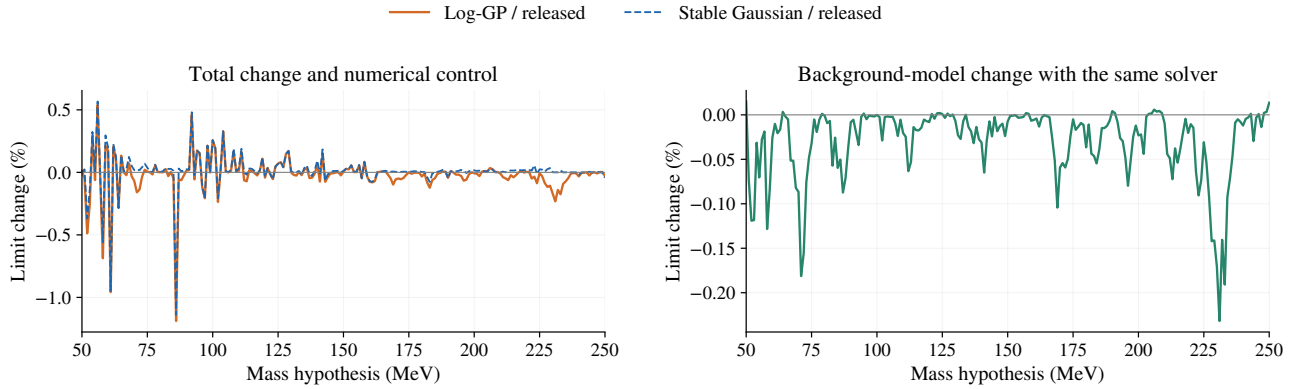


Figure 2: Separating model dependence from numerical differences. Left: direct log-GP and a stable re-solution of the Gaussian model, each relative to the release. Right: the two models evaluated with the same stable solver. The maximum Gaussian-control/release difference is 1.139%; the maximum same-solver distribution effect is 0.232%.

4 Signal-plus-background displays

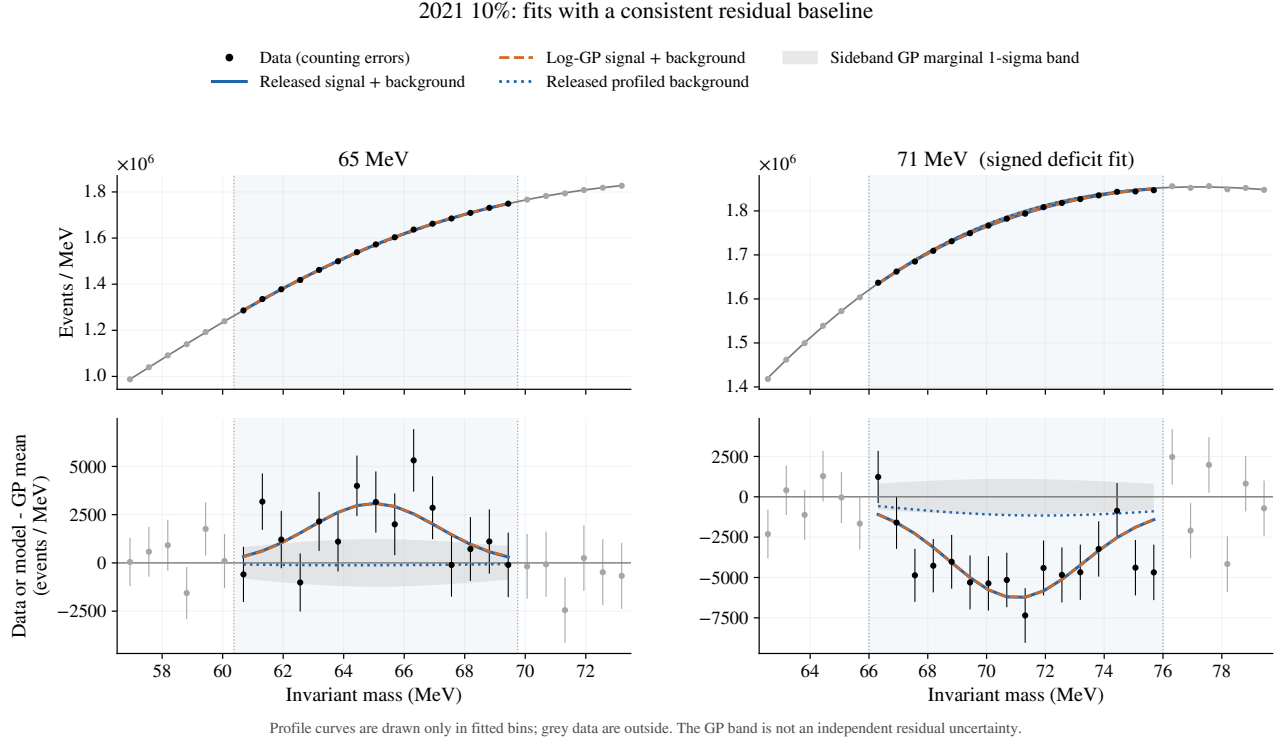


Figure 3: Fits at 65 and 71 MeV, including the signed deficit fit. Curves for a profiled background or total model stop at the actual fit mask; all residuals subtract the same sideband GP mean. This removes artificial joins between profiled and unprofiled baselines. The shaded GP band is a marginal background uncertainty, not an independent residual error or goodness-of-fit test.

The upper panels are dominated by the large smooth background. Subtracting a common GP mean makes the fitted signal and background displacement visible. The 65 MeV signal window and 71 MeV signal window belong to different sideband fits: their background predictions should not be interpreted as one global function with a signal added at several masses simultaneously.

At 71 MeV both profile treatments retain a signed negative fit with $r \simeq -4.02$. That sign is useful for diagnosing the extraction. The physical upper-limit denominator still uses the null fit, and the discovery convention in Eq. (4) gives $p_0 = 0.5$ for a deficit. A negative fitted yield is therefore not a physical negative signal or a discovery claim.

4.1 Selected fitted yields

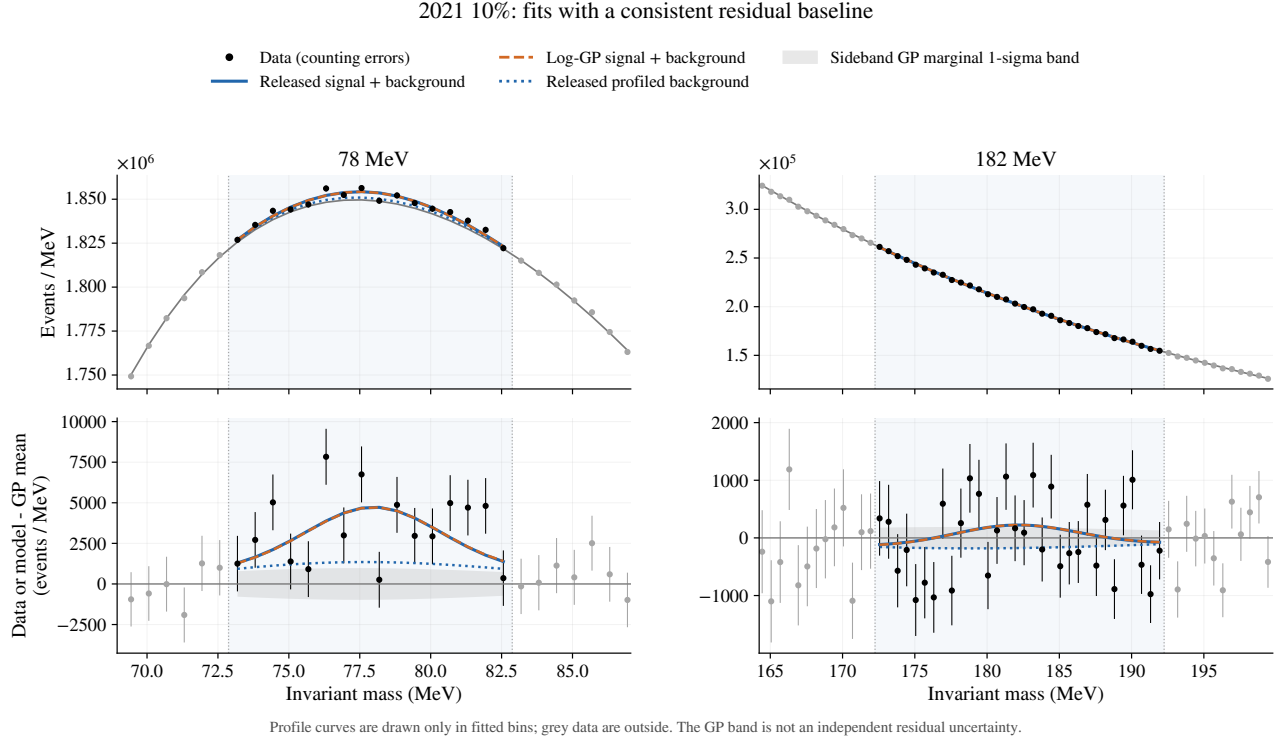


Figure 4: The same display at 78 and 182 MeV. The released fitted components are retained and the direct log-GP total is overlaid. The peak-dip behavior survives the change of background distribution. The v4.9.12.5 positive-injection studies demonstrate that neighboring signal entering the moving training region can induce negative responses; they do not identify the cause of the observed dip.

m (MeV)	Released profile		Direct log-GP		Fixed mean	
	$\hat{A}$	r	$\hat{A}$	r	$\hat{A}$	r
65	16,635	2.403	16,636	2.396	15,860	4.752
71	-27,504	-4.019	-27,497	-4.022	-35,380	-9.707
78	18,744	2.810	18,751	2.806	28,356	7.552
182	4,349	1.570	4,347	1.571	1,839	1.056

Table 2: Signed signal yields in the actual fit window (events) and signed likelihood roots. Negative values are extraction diagnostics; discovery uses $Z = \max(r, 0)$. The three columns of r are conditional on their respective background treatments.

At 78 MeV, fixing the background increases the fitted window yield from about 18,744 to 28,356 events; the fixed upper limit is consequently weaker despite its smaller formal error. At 182 MeV the fixed yield is smaller. Profiling changes the optimal weighting and background displacement as well as the uncertainty. A ratio of observed limits cannot isolate those effects.

5 Fixed-background observed extension

For the combination, the counts and signal counts per unit ϵ^2 from each dataset are concatenated and one shared coupling is fitted. Dataset resolution, normalization, and signal-window fractions enter that likelihood directly. Limits and p-values are not combined algebraically.

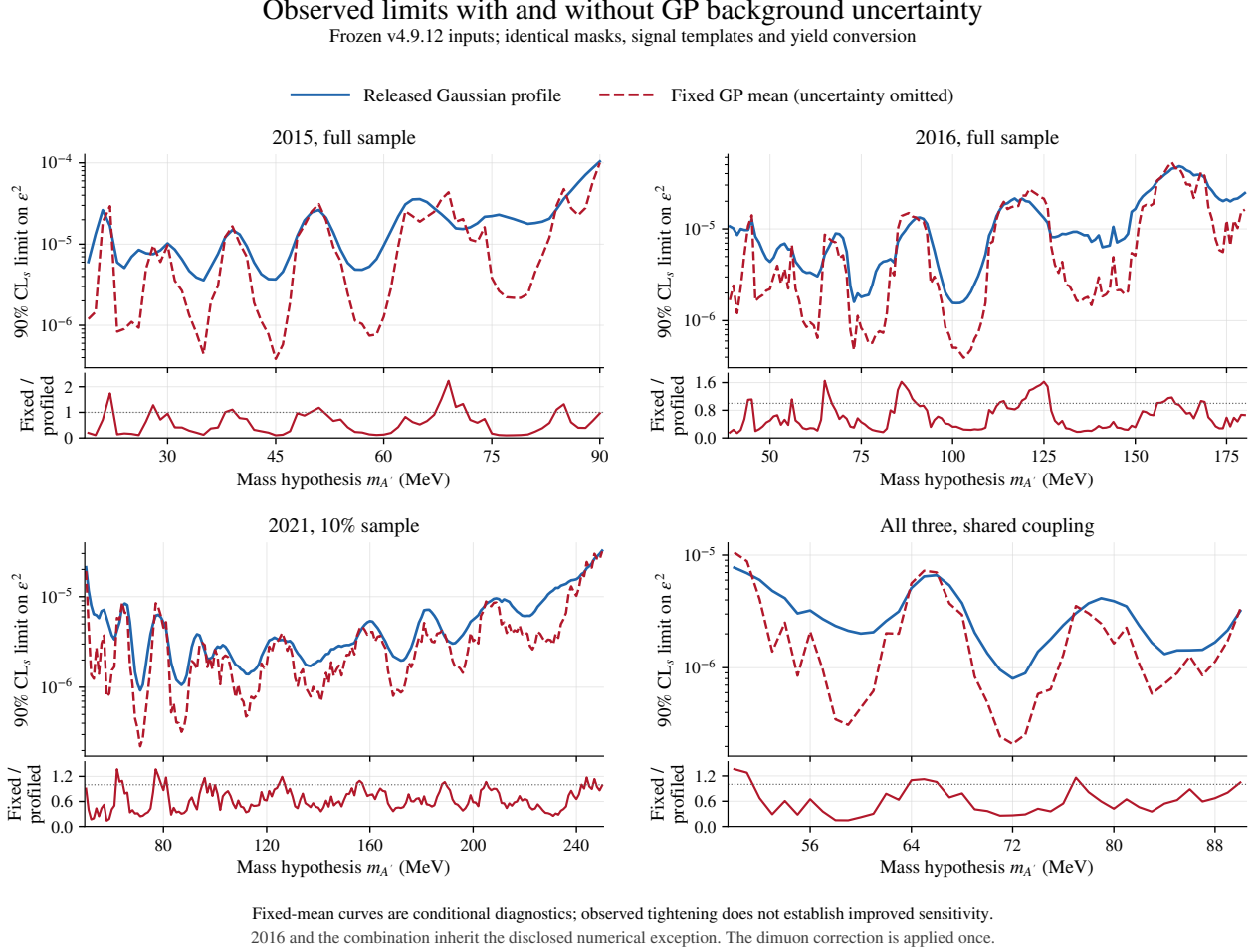


Figure 5: Observed 90% CL_s limits for full 2015, full 2016, 2021 10%, and the all-three shared-coupling fit. Ratio panels show fixed/released. Each panel has its own mass range and vertical scale. Fixing the mean changes both the uncertainty and fitted yield, so some observed limits become weaker.

Scope	Minimum ratio	Median ratio	Maximum ratio	Fixed smaller
2015, 100%	0.102	0.422	2.234	60/72
2016, 100%	0.141	0.522	1.653	114/142
2021, 10%	0.137	0.607	1.373	180/201
All three	0.146	0.595	1.361	34/41

Table 3: Fixed/released observed-limit ratios. Ratios describe the displayed conditional limits; a smaller value alone does not establish increased sensitivity.

5.1 Local asymptotic background-only p-values

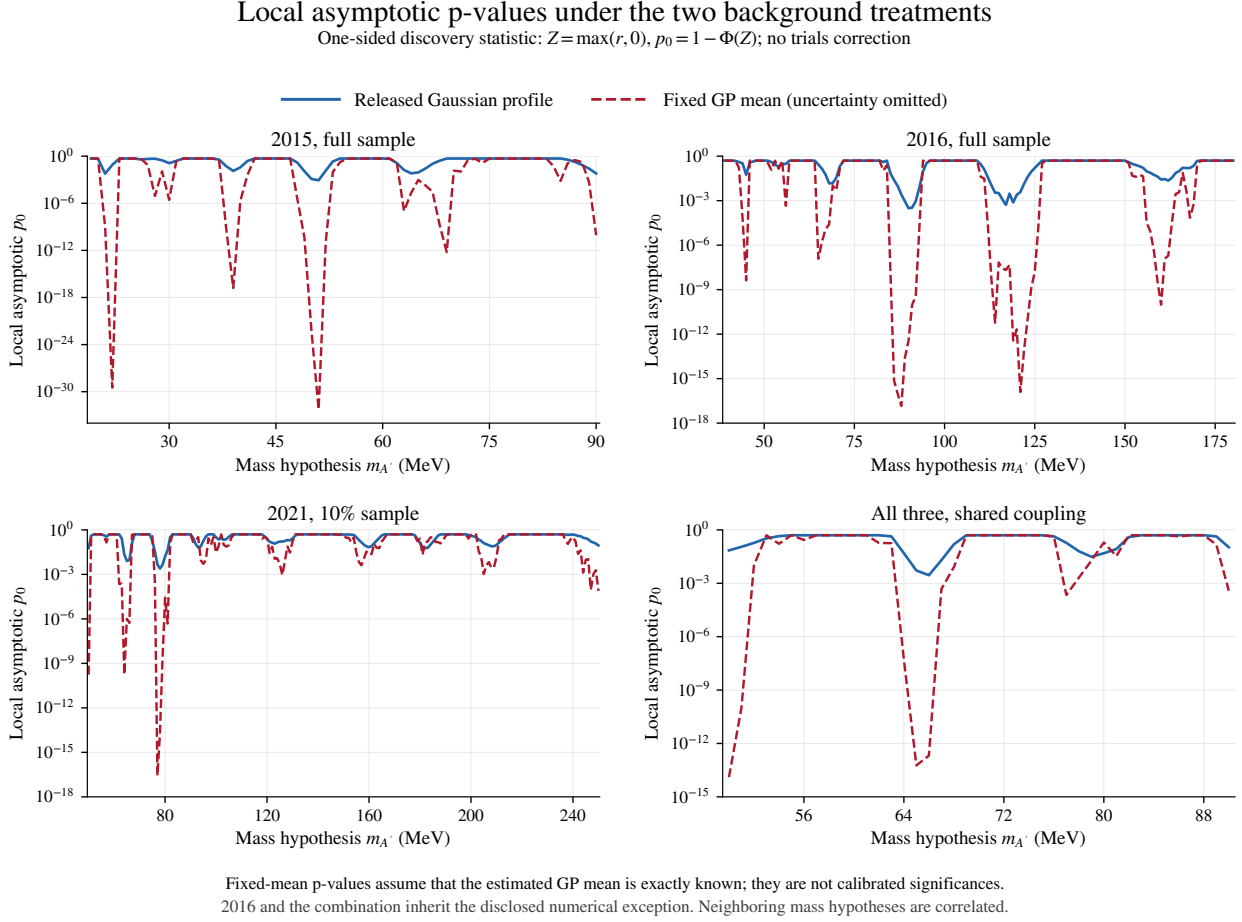


Figure 6: Local p_0 using Eq. (4). The fixed curves show the consequence of treating an estimated background as exactly known; their tiny values are not calibrated discovery evidence. Mass hypotheses are correlated. The 2016 state and the all-three curve retain the parent’s numerical exception.

Scope	Released Gaussian profile			Fixed GP mean		
	m (MeV)	Minimum p_0	Z	m (MeV)	Minimum p_0	Z
2015, 100%	51	8.46×10^{-4}	3.139	51	5.48×10^{-33}	11.906
2016, 100%	90	3.09×10^{-4}	3.424	88	1.43×10^{-17}	8.453
2021, 10%	78	2.47×10^{-3}	2.810	77	2.70×10^{-17}	8.378
All three	66	2.84×10^{-3}	2.765	50	1.27×10^{-14}	7.620

Table 4: Minimum local asymptotic p_0 in each scope. Fixed columns assume an exactly known GP mean; their quoted Z values are model-conditional transformations, not calibrated significances. No global correction or local rescaling is applied.

The minima occur at different masses for the two treatments in three scopes. A minimum local p-value is not the global p-value of the scan. Even at a fixed mass, the fixed-model asymptotic mapping can fail when GP estimation errors are omitted, as the 2021 toys below demonstrate.

5.2 Separating an observed fluctuation from conditional precision

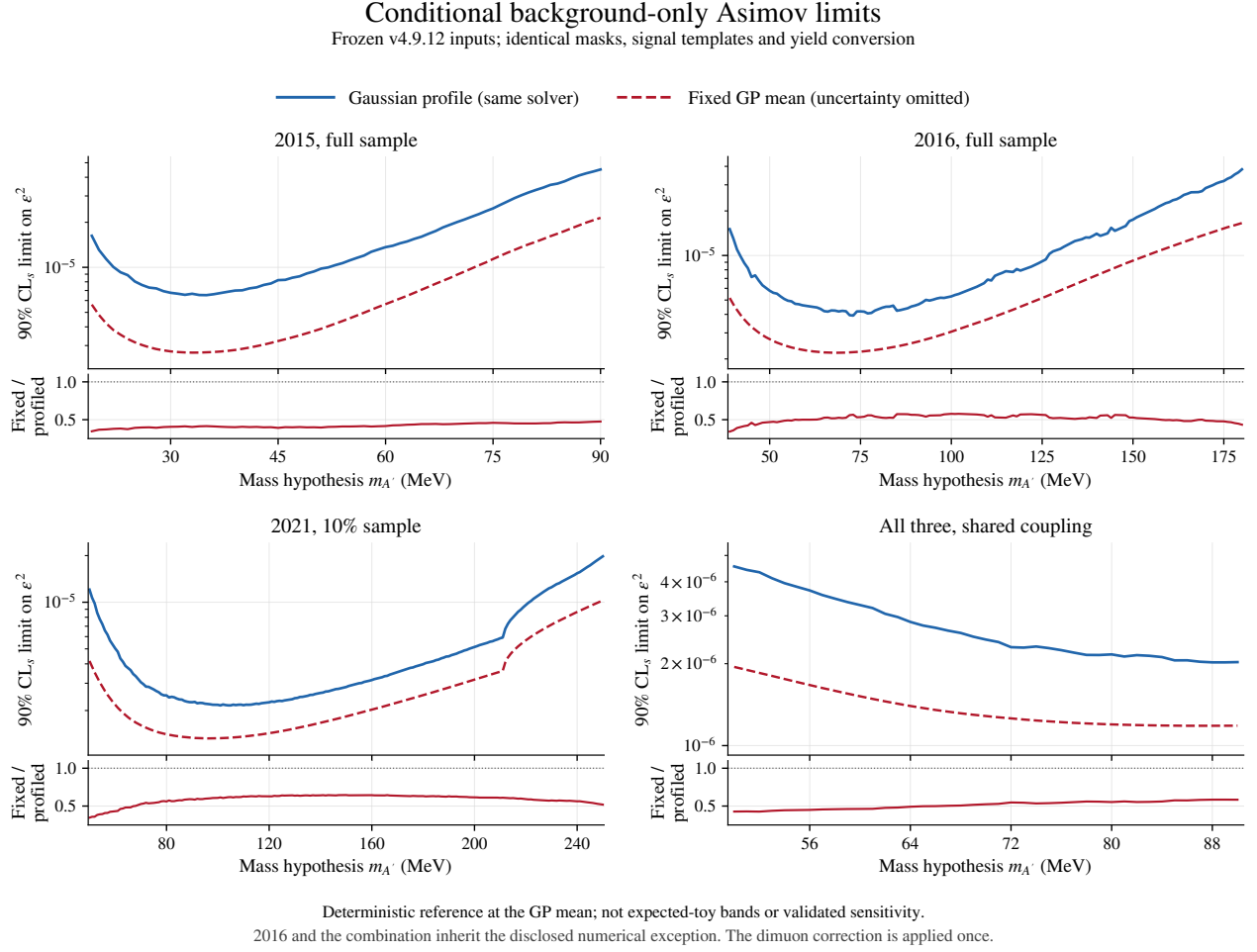


Figure 7: Deterministic background-only Asimov limits at the frozen GP mean, using the same stable solver for fixed and Gaussian-profiled backgrounds. These curves remove the observed signal fluctuation from the comparison, but retain the assumed background and constraint model. They are not expected-toy bands or a calibration of sensitivity.

The median fixed/profiled Asimov ratios are 0.412, 0.523, 0.613, and 0.524 for 2015, 2016, 2021, and all three. Thus a substantial conditional precision gain remains even when the observed fluctuation is removed. It arises from discarding the nuisance covariance: the relevant question is whether the background can actually be known to that precision.

A fixed-background likelihood is appropriate when independent information makes prediction uncertainty and bias negligible relative to the signal measurement. It is also useful as an ideal benchmark. Here the GP mean is estimated from finite, noisy sidebands. Passing injections generated from that same mean alone would test an ideal assumption, not establish it for the data. The following three ensembles separate those issues.

6 Signal injection and extraction

The protocol declares 500 spectra at each of 65, 71, 78, 100, 182, and 231 MeV, for $A_{\text{true}} = 0, 2\sigma_{\text{ref}}, 5\sigma_{\text{ref}}$. Here σ_{ref} is the reference profiled Fisher error, fixed before generation. The methods fit the same spectrum and physical yield. Each mass/strength has independent deterministic seeds; different strengths are not paired. These are partly data-selected diagnostic masses, not independent validation regions.

1. **Known-background control:** draw $n \sim \text{Pois}(b + Aw)$. The fixed assumption is exactly true. The profile fit still retains C .
2. **Conditional GP uncertainty:** draw $B = b + L\theta$, then $n \sim \text{Pois}(B + Aw)$. Both fits retain the nominal b, C . Nonpositive background vectors would be redrawn in full; none occurred.
3. **Retrained sidebands:** Poisson sample the complete saved smooth v4.9.12.5 truth plus the full signal, then refit the GP posterior outside the usual moving mask. Count-dependent training errors are recomputed; kernel coordinates remain frozen. Signal tails remain in training as the mask permits.

The third truth was fitted to observed data outside 60–86 MeV using the 66 MeV kernel. It is one selected GP-shaped spectrum, not an independently specified family of physical backgrounds. The first two ensembles test conditional uncertainty accounting; the third also tests posterior retraining. None includes hyperparameter reselection.

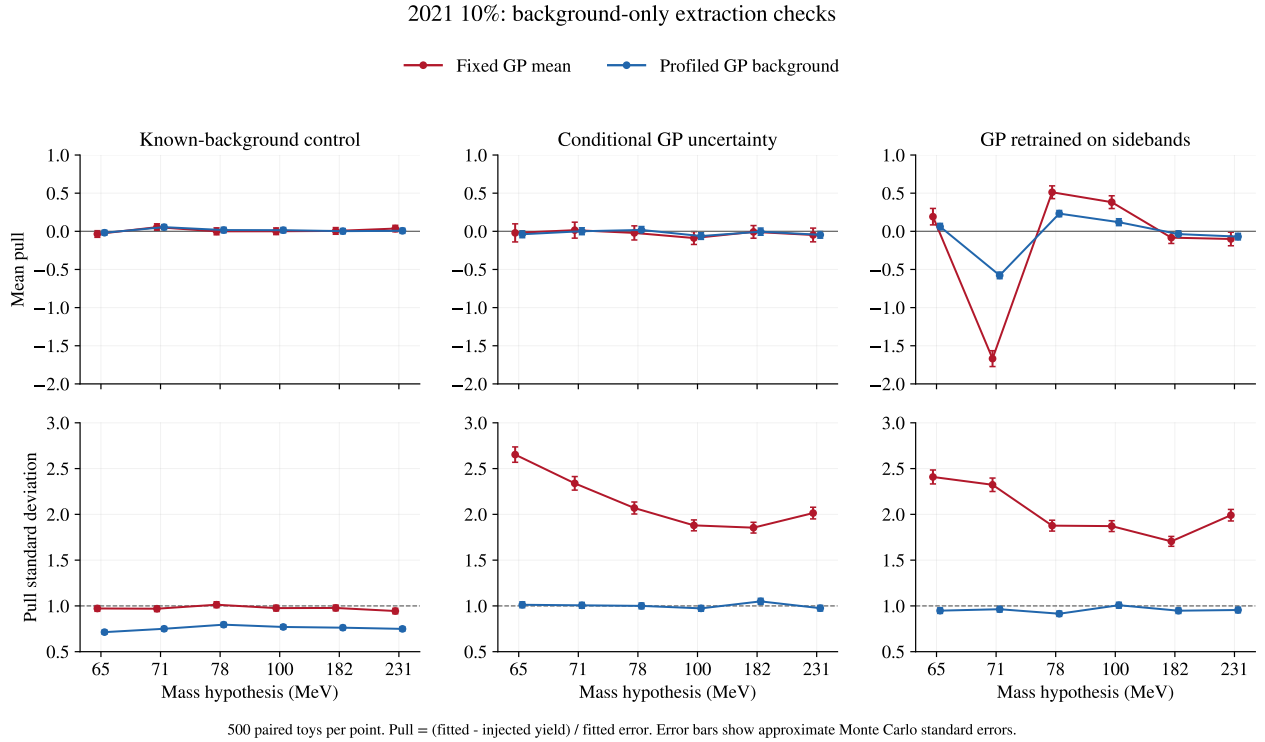


Figure 8: Background-only extraction. Pull is $(\hat{A} - A_{\text{true}})/\hat{\sigma}_A$. Masses are equally spaced categories here and in the next toy figures. Bars show approximate Monte Carlo standard errors. Fixed pull widths are 0.94–1.01 in the known-background control and 1.85–2.65 with GP uncertainty. Profiled widths in the latter are 0.97–1.05.

6.1 Bias and exclusion of the injected yield

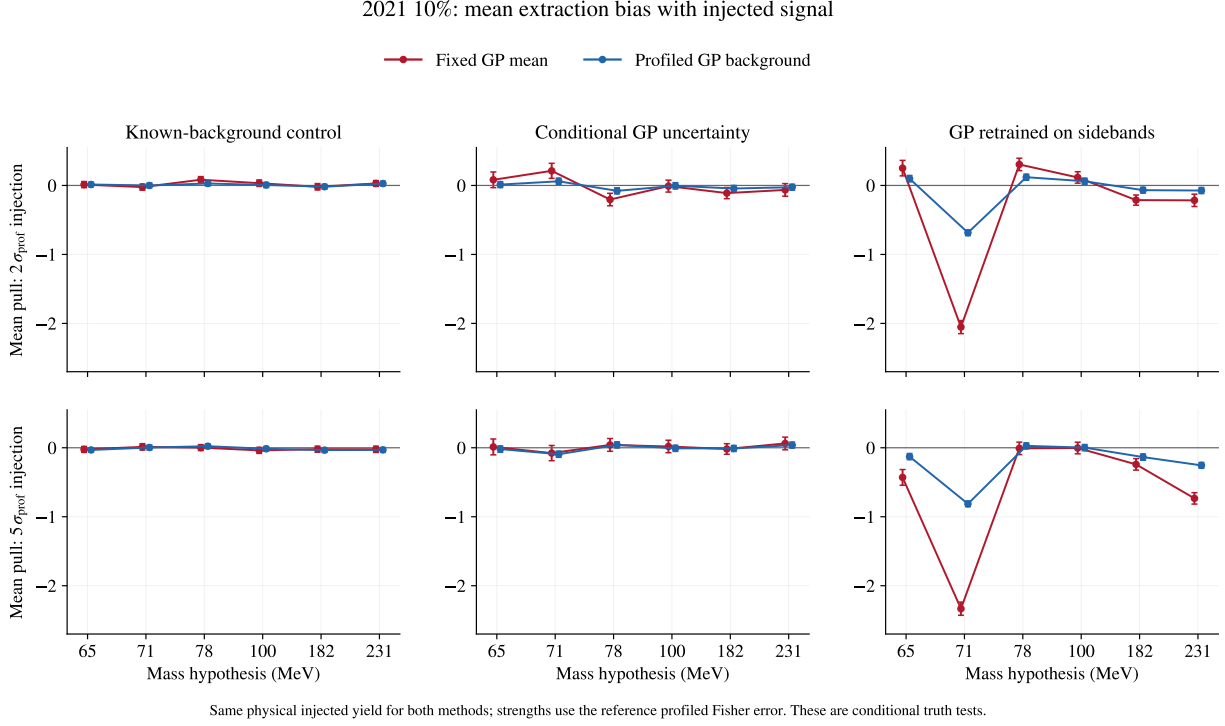


Figure 9: Mean pull for the two positive injections, using the same physical yield for both methods. The retrained 71 MeV truth produces bias in both fits. Bars are standard errors of the mean pull, not signal-yield uncertainties.

When the background is truly known, the fixed pull width is close to one and signal recovery works. The profiled fit is conservative in this deliberately ideal control: its covariance describes a source of variation absent from generation. Signal recovery alone therefore does not reject a fixed model.

When backgrounds vary according to the nominal GP covariance, fixed pull widths rise to 1.85–2.65. At zero signal, the nominal local $p_0 < 0.05$ criterion fires in 17.4–24.6% of fixed fits, versus 4.2–6.0% for the profiled fit. The fixed estimator can be nearly unbiased while assigning errors and p-values that are too small.

Refitting sidebands adds bias. At 71 MeV, the background-only mean pulls are -1.67 fixed and -0.58 profiled. At $5\sigma_{\text{ref}}$ they become -2.33 and -0.81 . Subtracting the background-only ensemble mean from the stronger injection mean gives a response of 92.9% fixed and 95.3% profiled. These are differences of separately generated ensemble means, not paired background-only/signal-plus-background differences. Reasonable incremental response can coexist with a biased absolute result.

6.2 Exclusion frequencies and current-method limitations

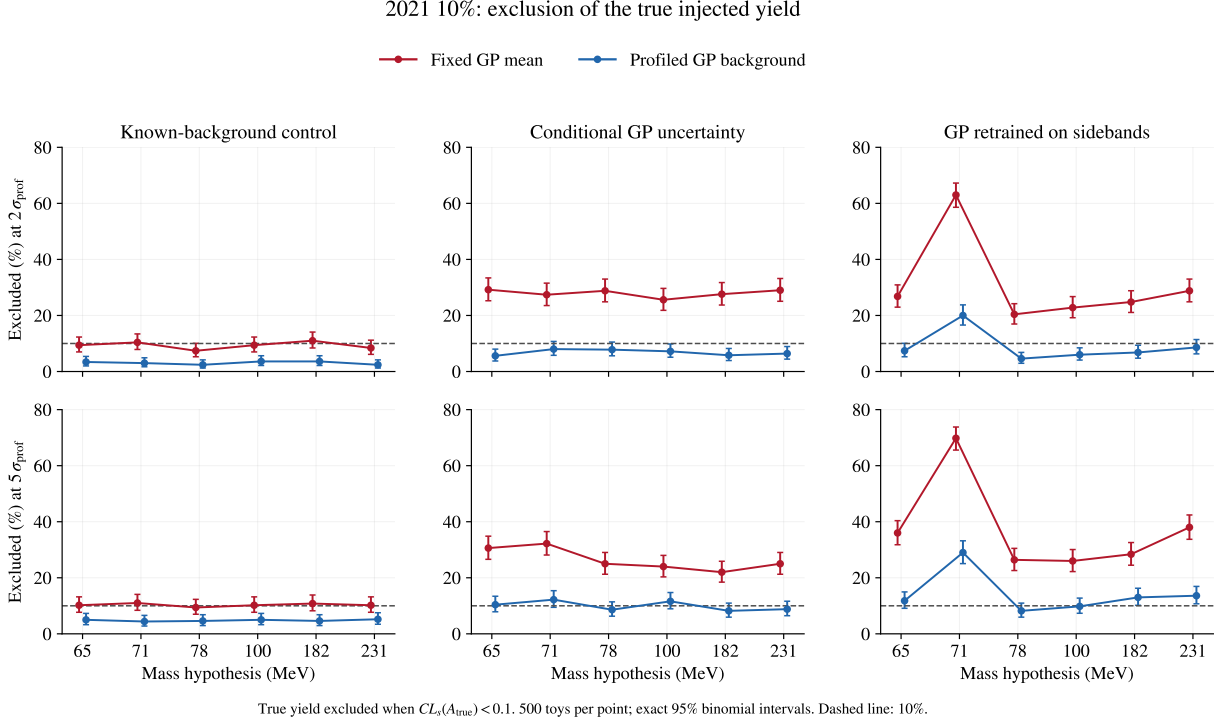


Figure 10: Fraction of toys with $CL_s(A_{\text{true}}) < 0.1$, with exact 95% binomial intervals. The 10% reference is the nominal upper-bound benchmark; CL_s can be conservative. At zero injection the physical bound would make this test uninformative, so only positive strengths are shown. These are conditional exclusion frequencies, not unconditional coverage estimates.

With known background, the stronger true injection is excluded in 9.4–11.0% of fixed fits. With GP uncertainty, that range rises to 22.0–32.2%, versus 8.2–12.2% profiled. In the retrained 71 MeV test, the stronger true yield is excluded in 349/500 fixed toys (69.8%; 95% interval 65.6–73.8%) and 145/500 profiled toys (29.0%; 25.1–33.2%). At $2\sigma_{\text{ref}}$, the fractions are 63.0% and 20.0%. The current profile treatment therefore also fails this particular conditional stress test. Good closure in the GP-uncertainty ensemble does not override it.

The v4.9.12.5 studies and this retraining test address related but distinct questions. The earlier coherent scans show that a positive feature can enter neighboring training regions and generate a deficit-like response. Here each injection is tested at its own mass and each mass has a separate ensemble. The observed dip is not attributed to one mechanism, and neither study supplies a global tail probability.

The agreement of Gaussian and direct log-GP profiling makes the Gaussian count approximation an unlikely explanation for the observed structure. The retraining failure instead motivates validation of the generating shape, training mask, and model-selection procedure. A positive-background profile alone does not address those choices. The frozen 2016 replay exception is a separate numerical qualification and is not resolved by the present results.

7 Can a local significance scale recover the apparent gain?

In the high-count, locally linear model, write $D = \text{diag}(b)$ and assume the prediction residual has zero mean and covariance $V = D + C$. The fixed-background estimate and its reported information are approximately

$$\hat{A}_{\text{fixed}} = \frac{w^T D^{-1}(n - b)}{I}, \quad I = w^T D^{-1}w, \quad \sigma_{\text{fixed}}^2 = I^{-1}. \quad (5)$$

Propagating the omitted covariance gives

$$\kappa^2(m) = \frac{\text{Var}(\hat{A}_{\text{fixed}})}{I^{-1}} = 1 + \frac{(D^{-1}w)^T C (D^{-1}w)}{I}, \quad \sigma_{\text{fixed,corr}} = \kappa \sigma_{\text{fixed}}. \quad (6)$$

This is a model-based variance correction, related to the distinction between likelihood curvature and sampling variance under misspecification [3]. The Gaussian profiled estimator instead uses the covariance in its weights:

$$\sigma_{\text{prof}}^2 = (w^T V^{-1}w)^{-1}, \quad \sigma_{\text{fixed,corr}} \geq \sigma_{\text{prof}}. \quad (7)$$

The inequality follows from generalized least squares for linear unbiased estimators under the stated V . It is not a theorem about biased, selected GP fits or arbitrary nonlinear models.

Propagating GP uncertainty removes the apparent precision gain

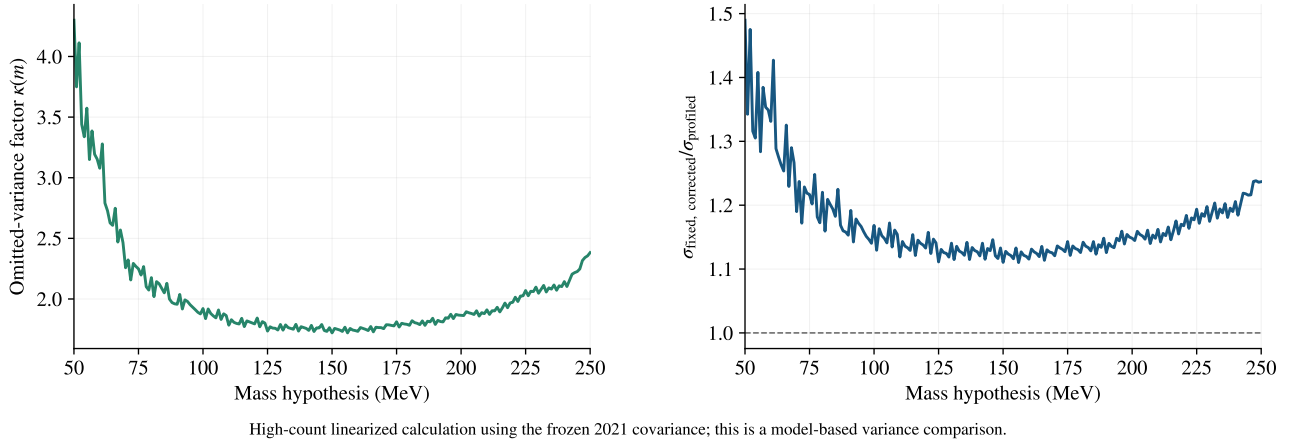


Figure 11: Covariance propagation across all 201 frozen 2021 masses. κ ranges from 1.72–4.30 (median 1.88). The corrected fixed/profiled error ratio has median 1.151 and range 1.110–1.490. The apparent precision advantage is lost within this zero-bias, Gaussian variance model.

Dividing the signed statistic by $\kappa(m)$ is consequently a useful diagnostic, but multiplying a significance by one universal number is not justified. Correcting a reported Z also does not recalibrate its associated CL_s upper limit. A valid limit would need a calibrated statistic and sampling distribution for each tested signal strength.

7.1 Held-out local calibration and the look-elsewhere effect

Before generation, Eq. (6) fixes the analytic scale. A separate empirical exercise estimates the mean and width of background-only r from toy IDs 0–99 at each mass and ensemble, then evaluates $(r - \bar{r}_{\text{train}})/s_{\text{train}}$ only on IDs 100–499. All three curves below use the same 400 held-out toys. Neither correction uses the observed statistic or is applied to the observed curves.

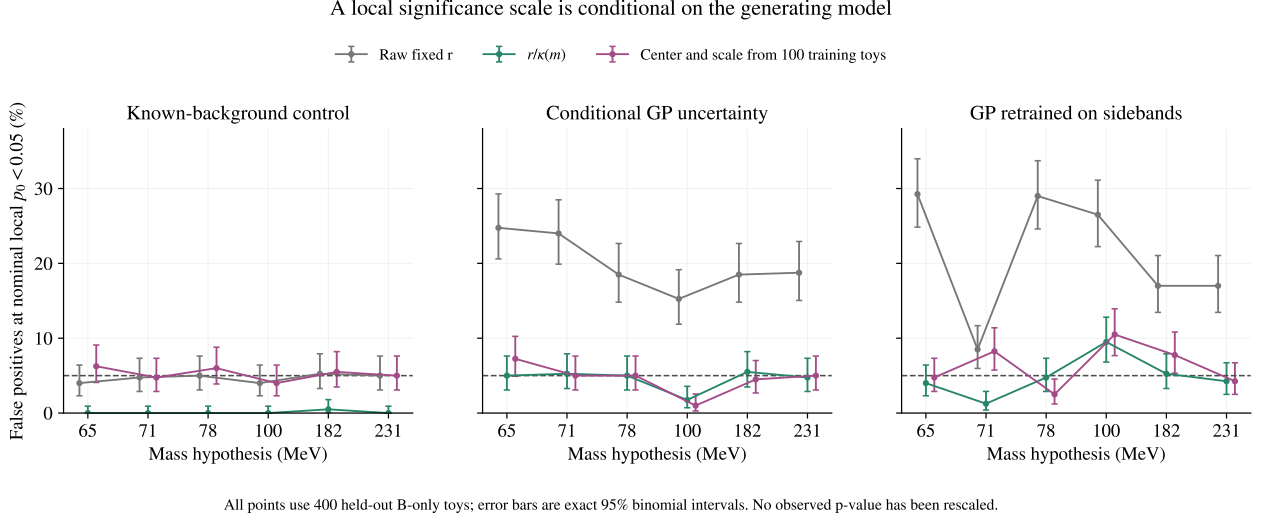


Figure 12: Nominal local $p_0 < 0.05$ false-positive fractions under the raw fixed statistic, the analytic variance correction, and the split-sample center/scale correction. Intervals are exact 95% binomial intervals conditional on the fitted training correction; they do not integrate over repeated training samples. Five hundred toys per coordinate cannot calibrate discovery-level tails.

For conditional GP uncertainty, the analytic correction changes the held-out false-positive range from 15.3–24.8% to 1.8–5.5%. In the retrained ensemble, the corrected fractions still range from 1.3–9.5%, and the empirical center/scale version ranges from 2.5–10.5%. A scale can address a variance error under its generating model, while bias, non-Gaussian tails, training uncertainty, and model dependence remain. In the known-background control the analytic scale overcorrects because the assumed GP variation is absent.

The global look-elsewhere correction addresses searching across correlated mass hypotheses [4]. It cannot repair an incorrect local null distribution. Local calibration must be established first; a coherent scan-wide ensemble or justified random-field calculation would then address the trials effect. No such correction is produced by these pointwise toys.

7.2 Recommendation

Retain fixed-background fits as known-background benchmarks and numerical diagnostics. The present evidence does not support their uncorrected limits as an improvement in sensitivity. Retain the current profiled result’s conditional status and the explicit 71 MeV retraining failure.

A defensible next model comparison would fit signal and a positive smooth functional background simultaneously, following the published HPS approach [5], or use a predeclared discrete-profile family [6]. Validate it on independent smooth truths and localized contaminations, with the complete training and selection procedure repeated. Background choices should be qualified before comparing favorable observed limits. Those models are proposals for a further study; the implemented alternative here is the direct log-GP profile.

A Numerical extraction results

Truth and mass	B-only pull width		Mean pull at $5\sigma_{\text{ref}}$		True yield excluded	
	Fixed	Profiled	Fixed	Profiled	Fixed	Profiled
<i>Known-background control</i>						
65 MeV	0.97	0.71	-0.02	-0.03	10.2%	5.0%
71 MeV	0.97	0.75	0.01	0.00	11.0%	4.4%
78 MeV	1.01	0.79	0.00	0.02	9.4%	4.6%
100 MeV	0.98	0.77	-0.04	-0.01	10.2%	5.0%
182 MeV	0.98	0.76	-0.02	-0.03	10.8%	4.6%
231 MeV	0.94	0.75	-0.02	-0.03	10.2%	5.2%
<i>Conditional GP uncertainty</i>						
65 MeV	2.65	1.01	0.01	-0.02	30.6%	10.4%
71 MeV	2.34	1.01	-0.08	-0.09	32.2%	12.2%
78 MeV	2.07	1.00	0.04	0.04	25.0%	8.6%
100 MeV	1.88	0.97	0.02	-0.01	24.0%	11.6%
182 MeV	1.85	1.05	-0.02	-0.01	22.0%	8.2%
231 MeV	2.01	0.98	0.06	0.04	25.0%	8.8%
<i>GP retrained on sidebands</i>						
65 MeV	2.41	0.95	-0.43	-0.13	36.0%	11.8%
71 MeV	2.32	0.96	-2.33	-0.81	69.8%	29.0%
78 MeV	1.88	0.91	-0.01	0.03	26.4%	8.2%
100 MeV	1.87	1.01	-0.00	0.00	26.0%	9.8%
182 MeV	1.71	0.95	-0.24	-0.14	28.4%	13.0%
231 MeV	1.99	0.96	-0.73	-0.26	38.0%	13.6%

Table 5: Conditional 2021 toy diagnostics. Each cell uses 500 spectra; the fixed and profiled methods are paired on the same spectra. The stronger injection is five times the reference profiled Fisher error, with the same yield for both methods. Exclusion means $CL_s(A_{\text{true}}) < 0.1$. Binomial intervals and both injection strengths are shown in the figures and retained in the CSV.

The full CSV also retains both injected strengths, fitted yields and errors, signed roots, exact binomial intervals, and every toy coordinate. Zero-signal exclusion counts are not interpreted as coverage evidence. In the CSV, `false_positive_fraction` denotes a false-positive fraction only in background-only rows; at positive injection it is rejection power.

B Reproducibility and checks

All 415 native observed GP prediction hashes match the v4.9.12 ledger. The exact archived runtime, native histogram coordinates, reviewed kernels, and conditioning conventions are retained. The 2016 exception is neither silently removed nor independently recertified. Work is isolated in the v4.9.13 derivative and does not modify the concurrent study’s files.

All 456 fixed observed limits passed independent scalar-fit and limit checks; 1,368 new observed/Asimov limits passed positivity, nesting, and root checks. The previous 201 fixed-2021 limits reproduce within 8.1×10^{-14} relative. The 54,000 extraction fits retain all 27,000 spectra, with no failed fits or nonpositive background redraws. There are 216 agreements between the true-yield exclusion shortcut and solved upper limits.

C Artifact and source record

The study root is `study_results/v4p9p13_background_profiling_20260905/`. Its README gives the reproducible commands. The LaTeX note and generated tables are under `note/`; standalone vector figures and numerical CSVs are grouped under `comparison/`, `observed/`, and `injections/`. `MANIFEST.sha256` records the deliverable identity, and the source ledgers record complete hashes. The principal inputs are:

- Frozen card, reviewed states, predictions and result columns: `study_results/v4p9p12_final_dataset_combinations_20260902/`.
- Original fitted components and typographic template: `study_results/v4p9p12_expanded_snapshots_20260905/`.
- v4.9.12.5 mechanism discussion: `study_results/v4p9p12_targeted_tail_refinement_20260905/note/added_sections.tex`.
- Saved smooth truth: `study_results/v4p9p12_2021_peak_dip_diagnostic_20toys_20260905/reverse_injection/derived/common_truth_and_signals.csv`.
- Validated direct-log-GP, Gaussian and fixed comparison: `study_results/background_profile_comparison_20260905/`.

The hash prefixes are `3944d4c71a453c6c` for the 2021 histogram and `502dbc232780c445` for the smooth truth. Verification uses the complete SHA-256 hashes recorded in the ledgers. The dimuon correction is applied once to displayed ϵ^2 curves above $2m_\mu = 211.316749$ MeV. Electron-channel raw values remain in the CSVs; p-values are unaffected by this conversion.

References

- [1] M. Frate, K. Cranmer, S. Kalia, A. Vandenberg-Rodes, and D. Whiteson, “Modeling Smooth Backgrounds and Generic Localized Signals with Gaussian Processes” (2017), arXiv:1709.05681.
- [2] G. Cowan, K. Cranmer, E. Gross, and O. Vitells, “Asymptotic formulae for likelihood-based tests of new physics,” *Eur. Phys. J. C* **71**, 1554 (2011), arXiv:1007.1727, version 3.
- [3] H. White, “Maximum Likelihood Estimation of Misspecified Models,” *Econometrica* **50**, 1–25 (1982), primary paper.
- [4] E. Gross and O. Vitells, “Trial factors for the look elsewhere effect in high energy physics,” *Eur. Phys. J. C* **70**, 525–530 (2010), arXiv:1005.1891.
- [5] P. H. Adrian et al. (HPS Collaboration), “Searching for Prompt and Long-Lived Dark Photons in Electro-Produced e^+e^- Pairs with the Heavy Photon Search Experiment at JLab,” *Phys. Rev. D* **108**, 012015 (2023), arXiv:2212.10629.
- [6] P. D. Dauncey, M. Kenzie, N. Wardle, and G. J. Davies, “Handling uncertainties in background shapes: the discrete profiling method,” *JINST* **10**, P04015 (2015), arXiv:1408.6865.